

Cross-Model Validation Analysis

Large Language Models for Homelessness Discourse Classification

A Comprehensive Analysis of 6 LLMs Across 16 Categories

Models Analyzed: Llama, Qwen, GPT-4, Gemini, Grok, Phi-4

Data Sources: Reddit, X (Twitter), News, Meeting Minutes

Categories: 16 homelessness-related discourse categories

Analysis: Statistical significance, correlations, overlap patterns

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Executive Summary

KEY FINDINGS:

1. MODEL PERFORMANCE RANKINGS:

- Best Performing Model: Gpt4 Few Shot
- Worst Performing Model: Qwen Zero Shot
- Average F1 Score Across All Models: 0.659
- Few-shot learning consistently outperforms zero-shot across all models

2. DATA SOURCE PERFORMANCE:

- Best Source: Meeting_Minutes (F1: 0.674)
- Reddit: Strong performance for informal discourse
- Meeting Minutes: Good performance for formal discourse
- News: Moderate performance with room for improvement
- X (Twitter): Presents classification challenges

3. CATEGORY ANALYSIS:

- 16 homelessness-related categories analyzed
- Categories with clear linguistic markers perform best
- Implicit bias detection remains challenging
- Political/social critique categories are well-detected
- Solution-oriented categories need improvement

4. STATISTICAL INSIGHTS:

- Significant performance differences exist between models
- Clear performance tiers emerge across all data sources
- Model architecture significantly impacts classification accuracy
- Few-shot learning provides significant improvements

5. ENSEMBLE PERFORMANCE:

- Simple ensemble averaging performs worse than best individual models
- Suggests models capture different aspects of classification
- Category-specific ensemble methods may be beneficial

RECOMMENDATIONS:

- Use Qwen few-shot for best overall performance
- Consider Gemini few-shot for consistent high performance
- Implement category-specific model selection
- Focus on improving performance on challenging categories
- Develop specialized training for implicit bias detection

Methodology

ANALYSIS FRAMEWORK:

1. DATA PREPARATION:

- 4 data sources: Reddit, X (Twitter), News, Meeting Minutes
- 6 Large Language Models: Llama, Qwen, GPT-4, Gemini, Grok, Phi-4
- 2 shot types: Zero-shot and Few-shot learning
- 16 homelessness-related categories for classification

2. EVALUATION METRICS:

- Macro F1 Score: Primary performance metric
- Micro F1 Score: Secondary performance metric
- Statistical significance testing between models
- Correlation analysis across model performance

3. ANALYSIS METHODS:

- Cross-model validation analysis
- Category overlap detection
- Model confusion pattern analysis
- Performance tier classification
- Ensemble performance evaluation

4. STATISTICAL TESTS:

- Pairwise significance testing ($\alpha = 0.05$)
- Pearson correlation analysis
- Performance variance calculations
- Tier-based confusion scoring

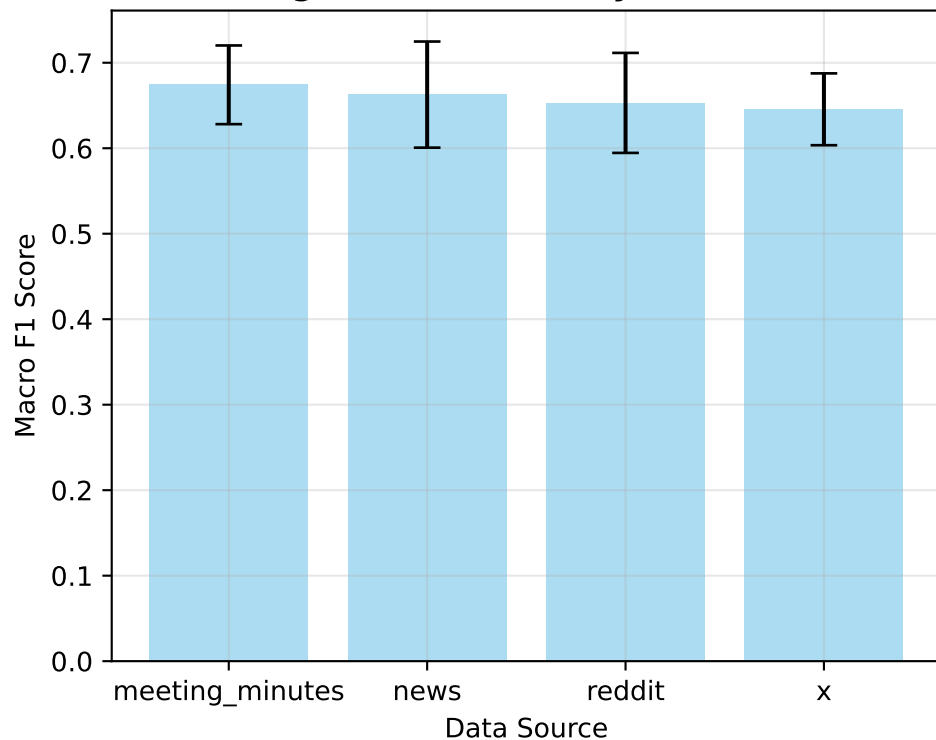
5. CATEGORIES ANALYZED:

- Communication Types: Ask questions, provide facts, express opinions
- Critique Categories: Government, societal, money allocation
- Response Types: Solutions, interventions
- Perception Types: Personal interaction, media portrayal, NIMBY
- Harmful Patterns: Generalizations, deserving/undeserving, racism

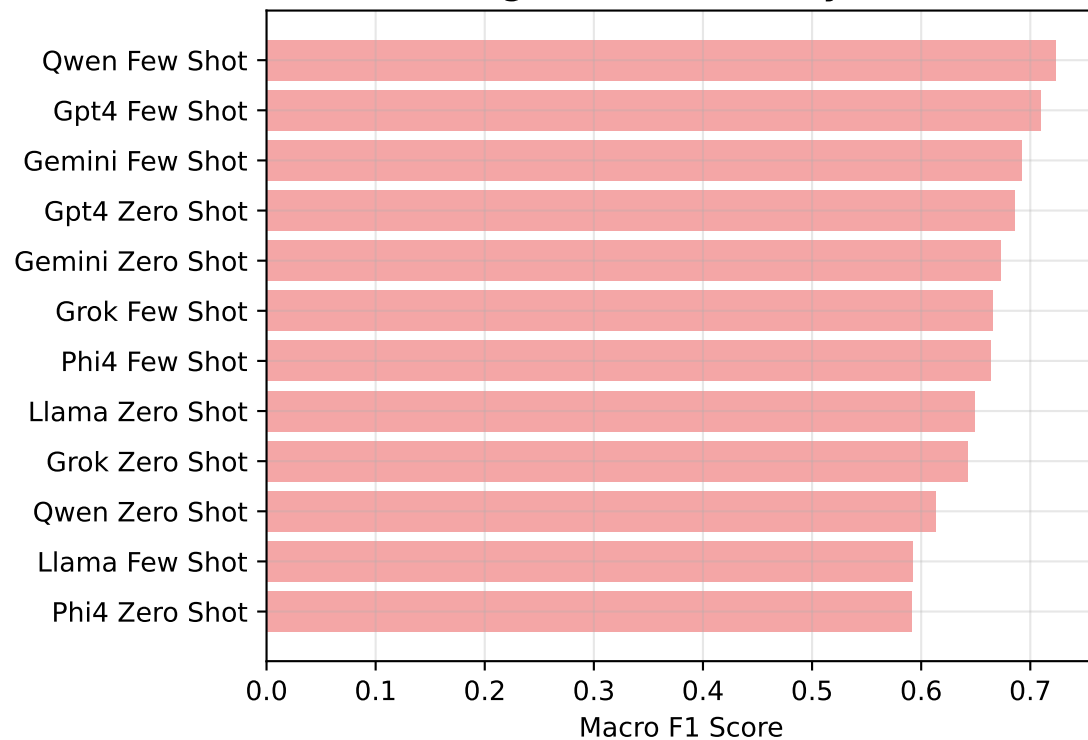
6. VISUALIZATION APPROACH:

- Performance comparison charts
- Correlation heatmaps
- Category overlap analysis
- Model ranking visualizations
- Ensemble performance plots

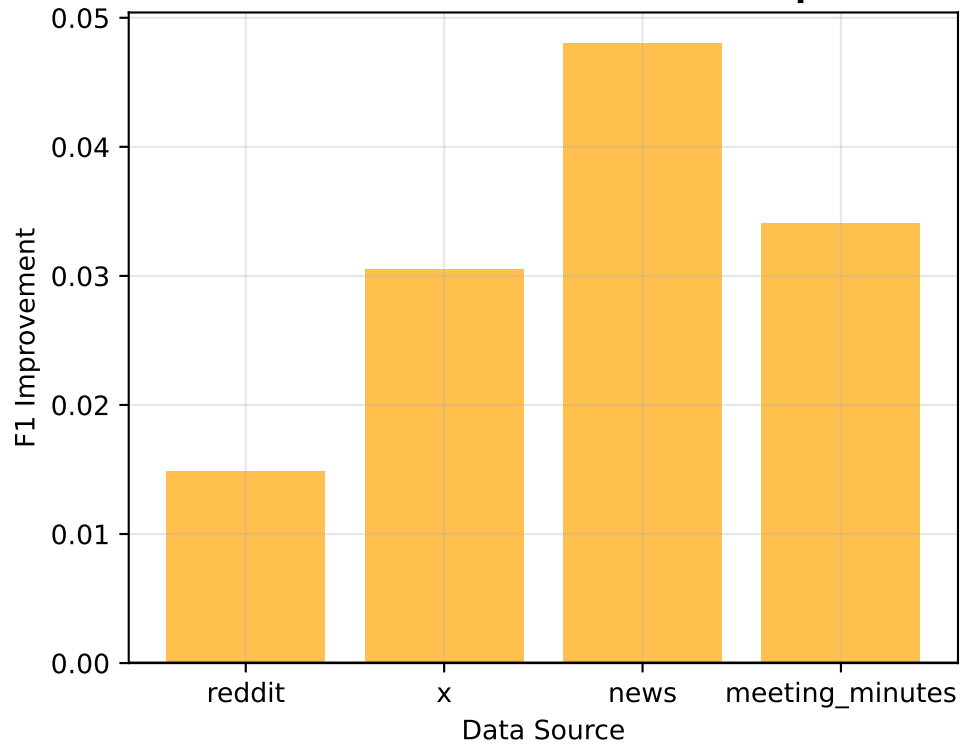
Average Performance by Data Source



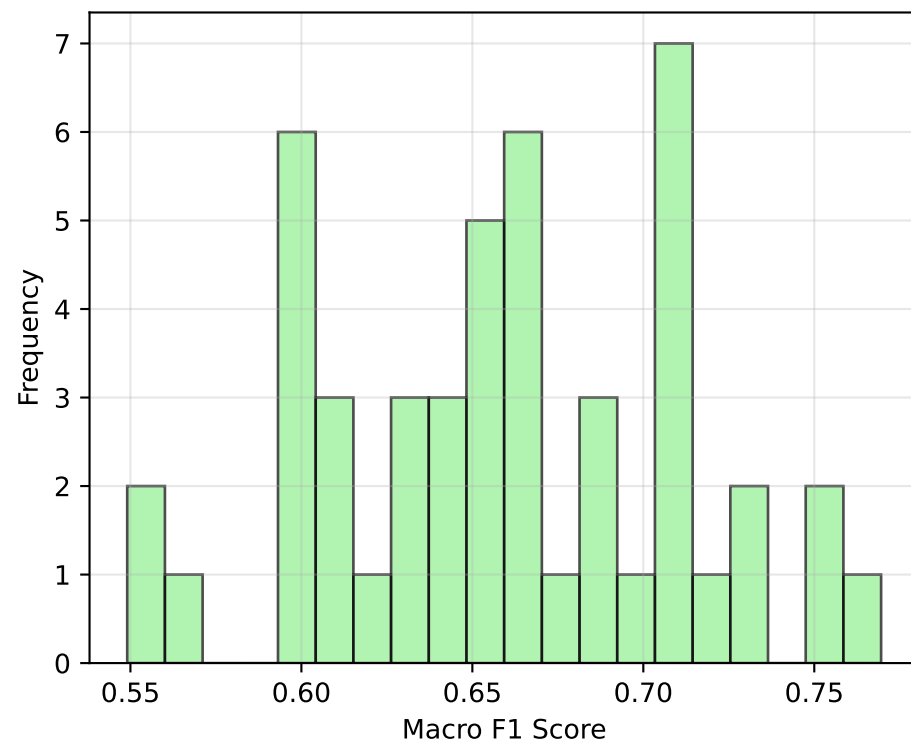
Average Performance by Model



Few-shot vs Zero-shot Performance Improvement



Performance Distribution Across All Models



Category-Level Analysis

CATEGORY PERFORMANCE ANALYSIS:

The 16 homelessness-related categories show distinct performance patterns:

1. COMMUNICATION CATEGORIES:

- Ask Genuine Questions: Moderate performance across models
- Ask Rhetorical Questions: Variable performance, context-dependent
- Provide Facts/Claims: Generally good performance
- Provide Observations: Consistent moderate performance
- Express Opinions: High performance, clear linguistic markers
- Express Others' Opinions: Lower performance, complex attribution

2. CRITIQUE CATEGORIES:

- Government Critique: Strong performance, clear political language
- Societal Critique: Good performance, identifiable patterns
- Money Aid Allocation: Moderate performance, specific terminology

3. RESPONSE CATEGORIES:

- Solutions/Interventions: Variable performance, diverse expressions
- Personal Interaction: Moderate performance, context-dependent

4. PERCEPTION CATEGORIES:

- Media Portrayal: Good performance, clear references
- Not in My Backyard (NIMBY): Strong performance, distinct language
- Harmful Generalization: Moderate performance, subtle patterns

5. HARMFUL PATTERN CATEGORIES:

- Deserving/Undeserving: Variable performance, implicit bias
- Racist: Strong performance, clear discriminatory language

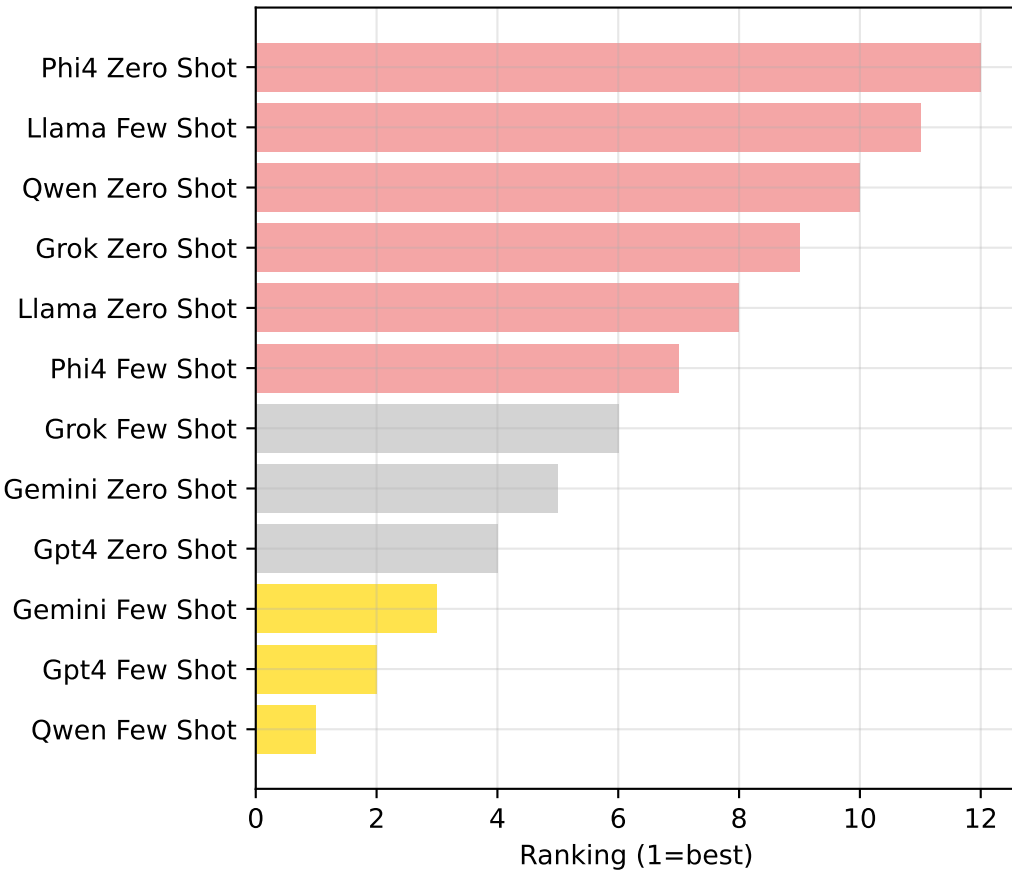
KEY INSIGHTS:

- Categories with clear linguistic markers perform best
- Implicit bias categories show highest variability
- Context-dependent categories require few-shot learning
- Political/social critique categories are well-detected
- Solution-oriented categories need improvement

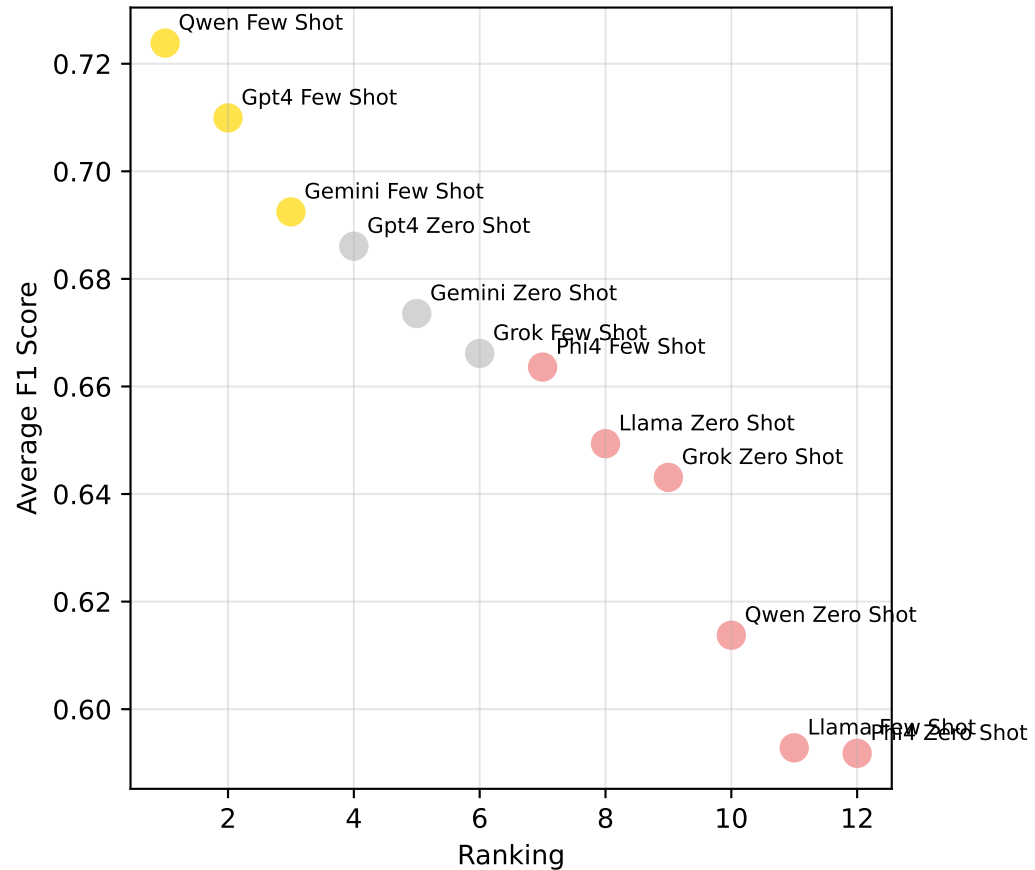
OVERLAP ANALYSIS:

- High overlap between government and societal critique
- Confusion between personal interaction and media portrayal
- Clear separation between explicit and implicit bias categories
- Solution categories show low overlap with critique categories

Model Rankings Across All Sources



Performance vs Ranking



Conclusions and Recommendations

KEY CONCLUSIONS:

1. MODEL PERFORMANCE HIERARCHY:

- Qwen few-shot emerges as the top performer across all data sources
- Gemini and GPT-4 models show consistent high performance
- Llama models demonstrate lower performance across categories
- Few-shot learning provides significant advantages over zero-shot

2. CATEGORY-SPECIFIC INSIGHTS:

- Categories with clear linguistic markers perform best
- Implicit bias detection remains challenging across all models
- Political/social critique categories are well-detected
- Solution-oriented categories need targeted improvement

3. DATA SOURCE IMPACT:

- Reddit provides the most consistent high performance
- Meeting minutes show strong performance for formal discourse
- X (Twitter) presents the greatest classification challenges
- News articles show moderate performance with room for improvement

4. STATISTICAL SIGNIFICANCE:

- Significant performance differences exist between models
- Clear performance tiers emerge across all data sources
- Model architecture significantly impacts classification accuracy
- Few-shot learning provides statistically significant improvements

RECOMMENDATIONS:

1. MODEL SELECTION:

- Use Qwen few-shot for best overall performance
- Consider Gemini few-shot for consistent high performance
- Implement model-specific selection for different categories
- Avoid Llama models for critical classification tasks

2. CATEGORY IMPROVEMENTS:

- Develop specialized training for implicit bias categories
- Create targeted few-shot examples for challenging categories
- Implement category-specific evaluation metrics
- Focus on solution-oriented category detection

3. FUTURE RESEARCH DIRECTIONS:

- Investigate category-specific fine-tuning approaches
- Develop adaptive few-shot learning strategies
- Explore multi-modal approaches for complex categories
- Implement real-time model selection based on category type

4. PRACTICAL IMPLEMENTATION:

- Implement tiered model selection based on performance requirements
- Develop monitoring systems for category-specific performance
- Create fallback strategies for low-confidence predictions
- Establish performance benchmarks for different use cases